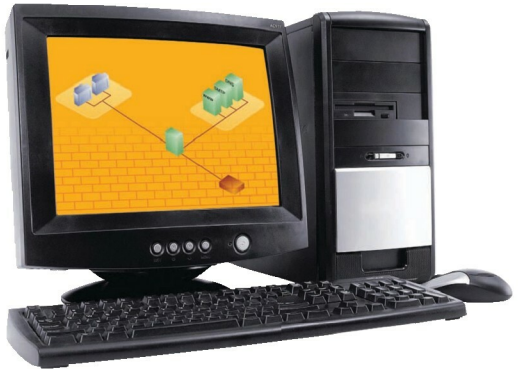




Visiting Netfilter

Let's create a small packet-filter firewall using the Netfilter packet filtering library available in Linux. The aim is to help network programming newbies understand socket buffers and Netfilter better.

I was writing a small firewall as my college project, when I got stuck with the *sk_buff* structure. This structure has changed in newer kernels, and most of the help I found on the Web was for the earlier version.

What is a socket buffer?

A socket buffer is the fundamental data structure in Linux Network Programming. Packets that are going out from and coming in to the machine all have their data stored in

this buffer. It is a double-linked list structure, which stores the protocols' PDU (Packet Data Units) for transmission and reception through the socket interface. It was designed to hold IP datagrams, but has been expanded to handle many other protocol PDUs.

You can look at the *sk_buff* structure members in the kernel source in the file *include/linux/skbuff.h*. The parts of the buffer are divided into general (maintenance of list), PDU headers (Ethernet, TCP, UDP, etc), memory, buffer, and IP datagram fields.